

Summary

BMI500 Profiling Exercise Report

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Environment: Local macOS (cluster SSH unavailable).

SLURM scripts: slurm/run_pbmc{3,6,10}k.sbatch

=== c) Coarse-grain (gtime -v) ===

pbmc3k: wall=0:21.42 user=14.25s sys=1.61s maxRSS=827.8 MB

pbmc6k: wall=0:16.29 user=16.20s sys=1.34s maxRSS=904.8 MB

pbmc10k: wall=0:30.64 user=28.18s sys=1.46s maxRSS=1696.4 MB

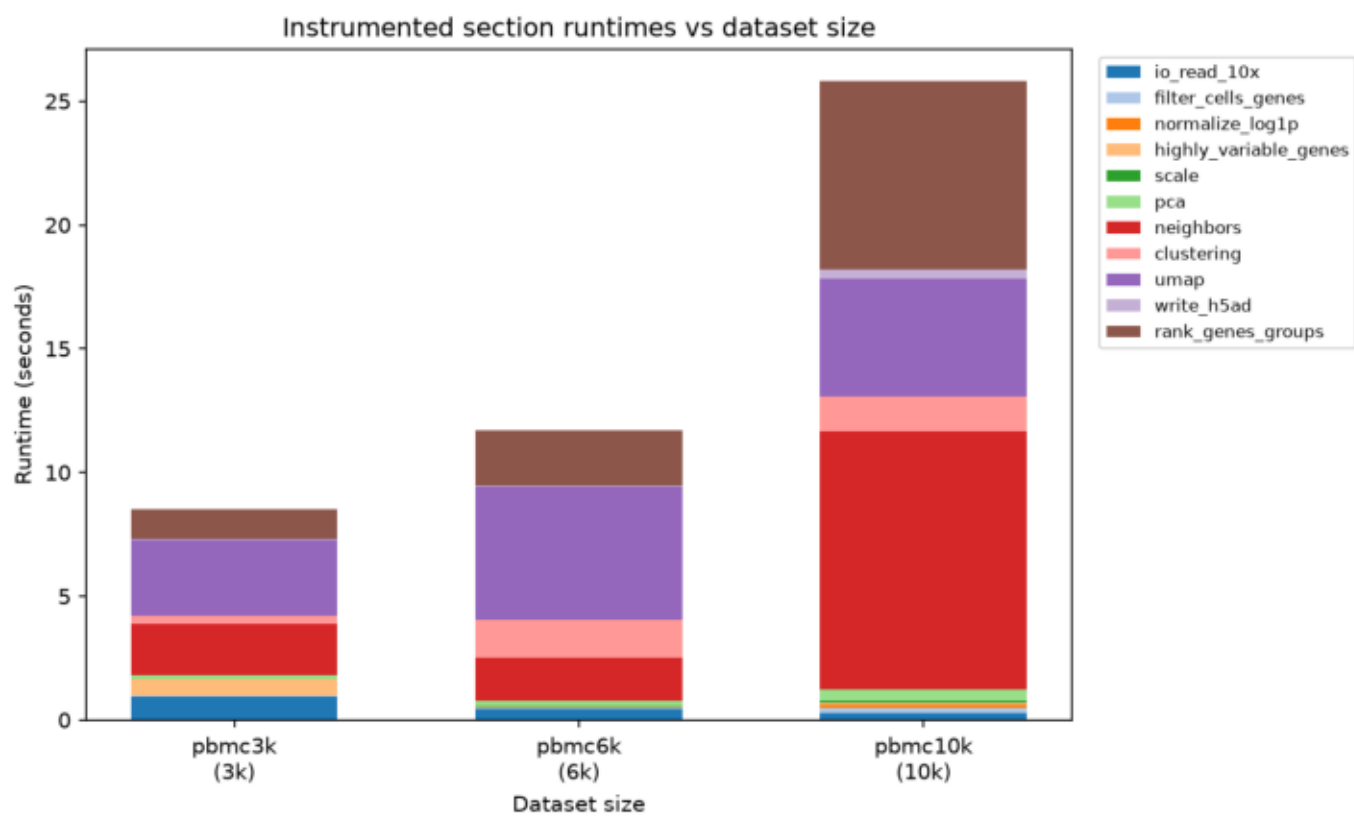
=== Instrumented section timings (s) ===

section	3k	6k	10k
io_read_10x	0.95	0.45	0.27
filter_cells_genes	0.02	0.02	0.22
normalize_loglp	0.01	0.02	0.11
highly_variable_genes	0.65	0.03	0.09
scale	0.03	0.05	0.07
pca	0.13	0.22	0.46
neighbors	2.07	1.74	10.43
clustering	0.33	1.50	1.39
umap	3.07	5.39	4.79
write_h5ad	0.05	0.07	0.36
rank_genes_groups	1.21	2.23	7.61

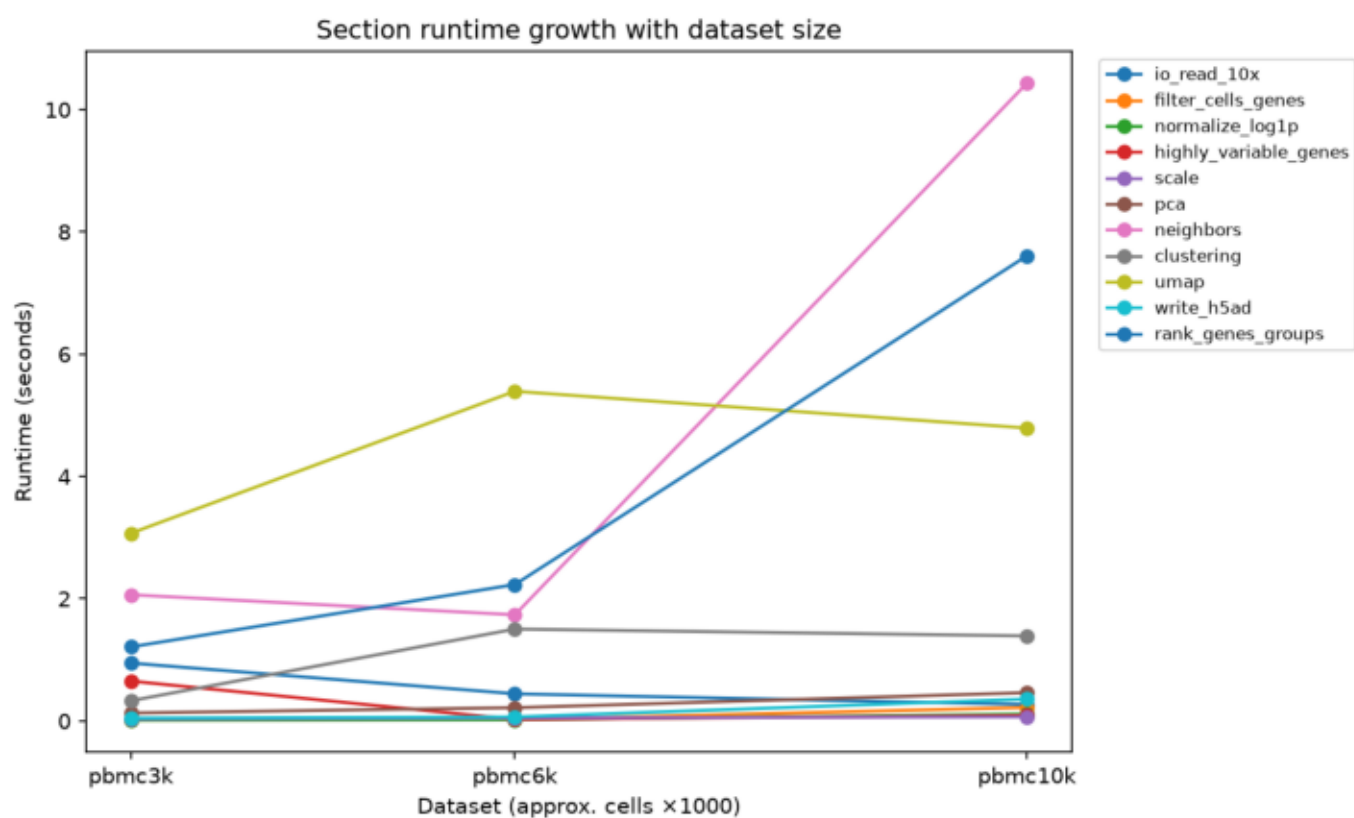
Major bottleneck (avg): neighbors (4.74s)

cProfile hot path: pandas.rank / Wilcoxon in rank_genes_groups

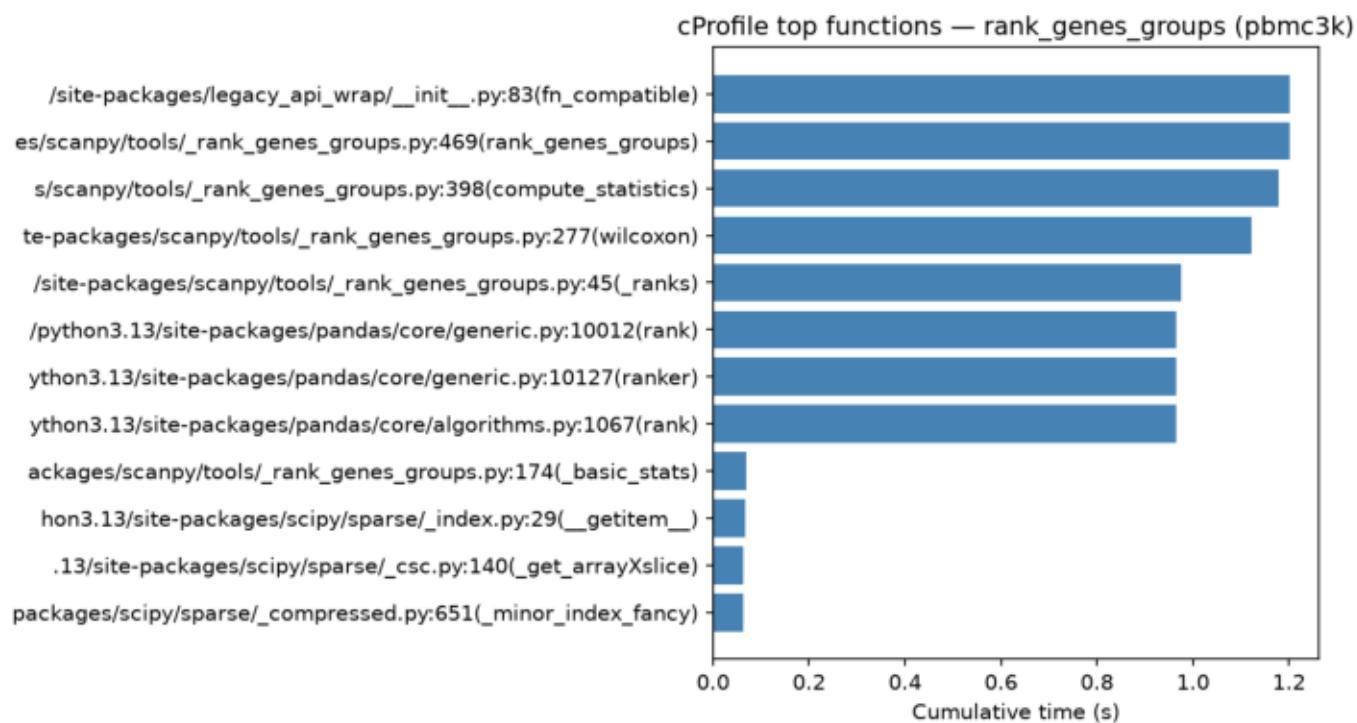
d) Stacked section runtimes vs dataset size



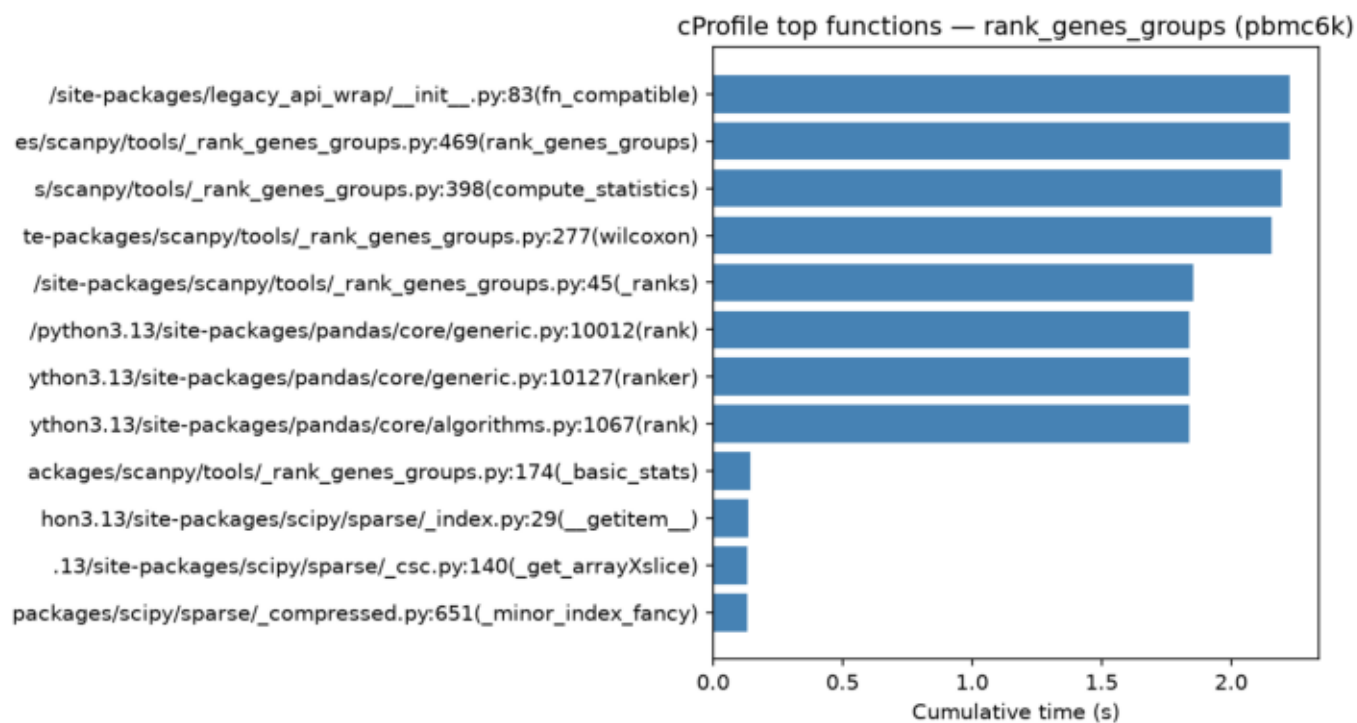
Section runtime lines vs dataset size



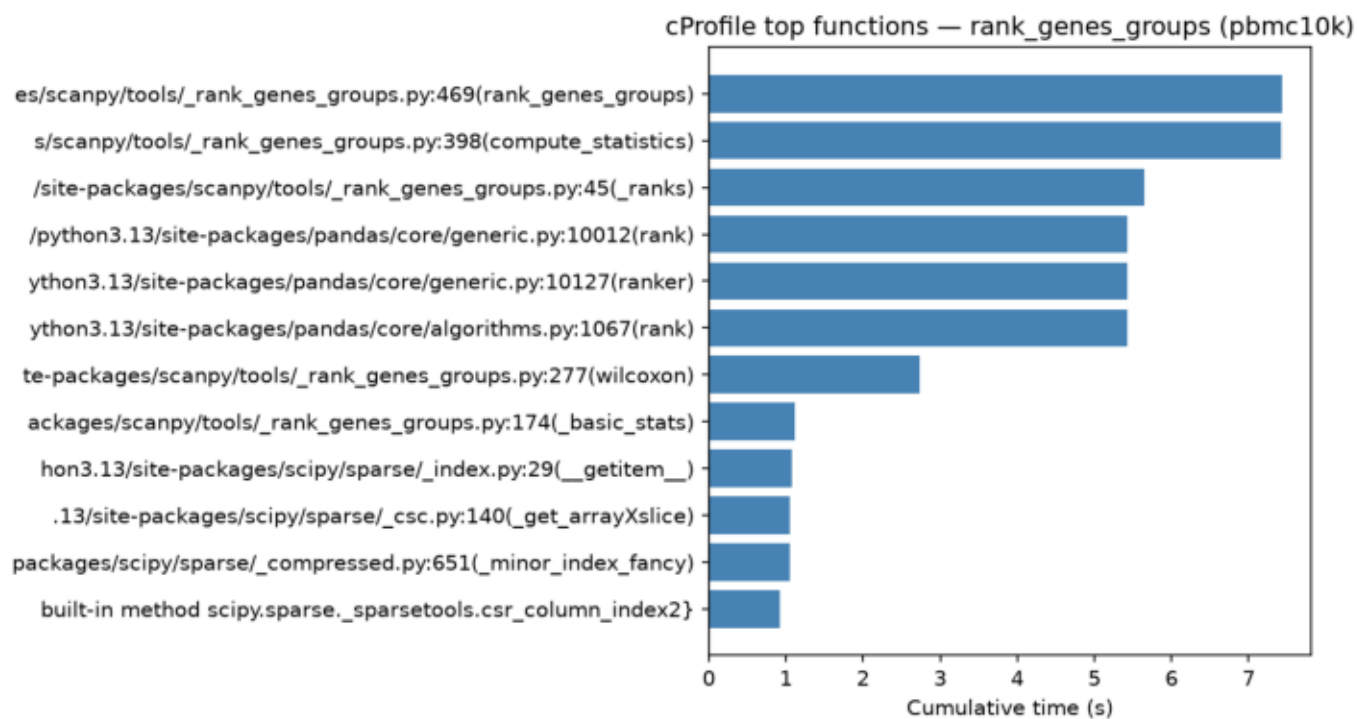
cProfile top - rank_genes_groups (pbmc3k)



cProfile top - rank_genes_groups (pbmc6k)



cProfile top - rank_genes_groups (pbmc10k)



Analysis

e) Bottlenecks and 20k extrapolation

Primary bottleneck: neighbors.
neighbors and rank_genes_groups grow with cells/clusters;
UMAP is also expensive.

section	pred20k	model
io_read_10x	0.13	power (y ~ x ^{-1.04})
filter_cells_genes	0.54	power (y ~ x ^{1.93})
normalize_loglp	0.26	power (y ~ x ^{1.67})
highly_variable_genes	0.01	power (y ~ x ^{-1.82})
scale	0.11	power (y ~ x ^{0.67})
pca	0.87	power (y ~ x ^{1.03})
neighbors	16.32	power (y ~ x ^{1.25})
clustering	4.29	power (y ~ x ^{1.25})
umap	7.08	power (y ~ x ^{0.40})
write_h5ad	0.80	power (y ~ x ^{1.62})
rank_genes_groups	18.01	power (y ~ x ^{1.49})

Sum predicted section times at 20k ~ 48.4 s

b) SLURM: run_pbmc3k/6k/10k.sbatch in slurm/
Note: leiden used locally (louvain build failed on macOS).